

Research Methods for People in a Hurry

An introduction to research methods in computer-human interaction, psychology, and human factors

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ABSTRACT

Research methods are a foundational part of an education in human-computer interaction. This education is not always provided, and what training is available may not always be focused on the most relevant topics for this diverse field. This course aims to survey research methods relevant to psychology, human factors, computer science, and engineering. This course is suitable for those with limited background in research, and can benefit those with substantial experience. This course aims to provide exposure to many relevant topics and inspire attendees to delve deeper and hone their craft with this course as an introduction.

CCS CONCEPTS

• **Applied computing** → **Education**; **Psychology**; • **Human-centered computing**;

KEYWORDS

Research Methods, Education, Professional Training

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1 BENEFITS

Research in human-computer interaction, human factors, usability studies, and psychological science requires a wide-ranging set of knowledge that is continuously evolving. Unfortunately, this knowledge is not always well taught as part of the curricula in the diverse disciplines of the CHI community. This course is intended to provide both a survey of research methods based in the experimental psychology and human factors traditions, and to open a forum for discussion of research methods in human-computer interaction. We will discuss the use of tools including simulation, behavior analysis, psychophysiological sensing, and how to design studies to properly use these techniques, see Figure 1. For attendees with little exposure to research methods, this course is intended to introduce

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Figure 1: Simulation in virtual reality can be a powerful tool for research in HCI, human factors, and usability investigation. Combining behavior analysis and psychophysiological measures of arousal such as heart rate and skin conductivity can yield a large amount of high-quality data that can support research designs that take into account multiple factors (triangulation).

the key tenets of research in human-computer interaction, human factors, and psychology. This is not intended to be a lecture, but a discussion: based on the knowledge level of participants, and their specific interests, the discussion can be tailored to focus on more advanced topics if the attendees have substantial knowledge of the material under discussion. Attendees with significant knowledge of research methods and/or statistics can contribute to the discussion and assist those earlier in their research careers to build foundational knowledge in this critical area.

2 INTENDED AUDIENCE

This course is intended for those relatively inexperienced in research and is suitable for undergraduate or graduate students in any discipline that interfaces with human-computer interaction, human factors, or psychology. This is especially important training for students coming from computer science or engineering departments where research training is not necessarily a part of the curriculum. More senior students or professionals with experience in this area can still benefit from attending this course to learn new techniques, deepen their knowledge, or contribute to teaching others. Those in teaching or mentoring roles in academia or industry can benefit from attending this course to learn techniques for teaching research methods and experimental design.

3 PREREQUISITES

There are no prerequisites for this course. Suggested readings are listed to get ahead on understanding the concepts that will be discussed in this course.

4 CONTENT

This is intended to be a two-unit (150 minute) interactive course.

A brief introduction to various types of research.

- Experimental Research: you get to manipulate things, but only one at a time.
- Observational Research: (Don't believe me? Just watch. 📺)
- Quasi-experimental research: kind of a middle-ground where you don't necessarily have random assignment or systematic manipulations, but the experimenters do influence participants or the environment.

Quantitative and Qualitative Research.

- Quantitative: things that can be counted and tabulated, ... like survey results, exam answers, task performance, reaction time, and wiggles of the EEG traces.
- Qualitative: descriptive information, like interviews and stories that can be analyzed in a more holistic way. In this area are traditions including observational and embedded research and methods such as grounded theory.
- Survey Research can exist under both qualitative and quantitative models. Surveys rely on asking good questions and not priming respondents (in which case you would get poor quality responses, and worse, you might not know why the data is bad).

Variables and types of data.

- Categorical data: things that are discrete, like yes or no answers to a question, or the number of participants in a CHI course (you can't have half a person take this course).
- Continuous data: things like measurements that are not in discrete points (sometimes called ratio or scale data). An example is mass: you could in theory have an infinitely fine scale that measures to the nanogram, or smaller.
- Choosing variables is a bit of an art and a science, and we will discuss how to determine what information to collect. More data is not necessarily better, as it allows for fishing for results—not a good thing! 🎣
- We will discuss how to design an experiment with statistical analysis in mind – and how to know what statistical tests to use. And why a simple design is probably the best choice, although there are reasons to use complex designs and advanced analysis techniques.

Validity, in some depth 📊.

- What are the various types of validity? (See Shadish, Cook, and Campbell, for a deep discussion of this)
- What are threats to validity (BE MINDFUL OF VALIDITY HAWK 🦅)?

The Arrow of Causation ➡ and causal modeling.

- Experimental research allows you to establish that A causes B. Well, ... sometimes.

- Observational research cannot really demonstrate causation: it can just provide evidence that C and D occur together. But you can perhaps make a claim that C has to come before D, if there are good reasons ... like birds are real because they existed before the CIA was founded (please pardon our leaning into the meme).
- Modeling causal chains is a challenge. We will do some work on modeling but as designs get more sophisticated this gets increasingly difficult. That's a reason to keep research designs simple!
- Mediation and moderation will be briefly discussed, but mediation analysis is beyond the scope of this course. That said, maybe this will whet your appetite to learn regression and statistical analysis techniques.

Designing a study.

- How to develop and select research questions 📝
- Selecting hypotheses (and what is a proper hypothesis anyway)? 🧠
- Between-participants, within-participants, and mixed designs. 🧑‍🔬

Research Ethics.

- What is ethics review, and why do we have it? If you want to go down that dark rabbit hole, look up the Tuskegee syphilis study, MKUltra, the Stanford Prison Experiment, or the commentary around the Facebook emotional manipulation studies conducted by Hancock et al. which is not actually unethical research as executed.
- What to do in case of adverse events and things on the ethics line - like participants suffering simulator sickness or telling people that deep down they have unconscious biases (see Implicit Association Test).
- How to deal with ethics review (IRB/REB) and the many rules. And the surprising things the review boards don't bother with (like art projects that can actually hurt people). Or the lack of oversight outside of academia (again, take a look at the commentary around Facebook's research on emotion manipulation).
- It should be obvious that research fraud is bad, but it does occur (see the sad stories of Diederik Stapel and Daryl Bem). While we don't expect anyone to be a real fraudster, how can we avoid even the minor scale p-hacking, 'fishing' for results, changing the hypotheses after data collection, and nudging our data to get results that get published but aren't really the truth ... ?

Questionnaires and surveys.

- We will design a survey to evaluate this workshop as a group exercise!

5 PRACTICAL WORK

This course comprises a guided discussion on research methods, how various methods can be applied in human-computer interaction contexts, and challenges in conducting research with human participants in laboratory, ecological, and virtual settings. This course can be taught in a virtual lecture and discussion format or

an in-person group discussion. In small groups, participants will discuss research topics, operationalize them as observational research, experiments, or quasi-experiments, and analyze the results. We will collaboratively develop a questionnaire to evaluate this workshop as an entire class group exercise.

6 INSTRUCTOR BACKGROUND

Dave B. Miller is a Graduate Faculty Scholar and Preeminent Postdoctoral Fellow in the department of Industrial Engineering and Management Systems at the University of Central Florida. Previously, he was a postdoctoral fellow at Cornell University and Cornell Tech. He earned his doctorate in Communication from Stanford University in 2019. His dissertation research focused on moral decision making and human-computer moral conflict in the frame of partially-automated driving. He earned a master's degree from the Gallatin School of Individualized Study at New York University and a bachelor's degree in human-environment relations and psychology from Cornell University. He has taught Psychology of Human-Technology Interaction at Stanford University and Ubiquitous Computing at Cornell University and workshops on human-computer interaction, research methods, and human-centered design at Aalto University, New York University, and the University of Central Florida. This course, Research Methods for People in a Hurry has been previously taught virtually, with attendees from UCF, NYU, Tufts University, Cornell University, and Brown University. Shaun Wallace is a doctoral candidate in the department of Computer Science at Brown University. Previously he was a Research Intern at the Adobe Creative Intelligence Lab, in addition to numerous industry roles as a software engineer. His dissertation research focuses on developing and deploying asynchronous collaboration systems in the wild to maintain evolving information. These systems provide paid and unpaid users with individual random examples they can use immediately to motivate them to interact continuously with and contribute new information. He earned a master's degree in Computer Science from Brown University, a second master's degree in Music Technology from the University of Limerick, and his bachelor's degree in Management Science and Information Systems from the University of Rhode Island. He has given several guest lectures in Human-Computer Interaction courses and applied these methods within several industry positions. A lecture of similar course material can be viewed here: <https://www.youtube.com/watch?v=MOtAEPJYp1w>

7 RESOURCES

While no readings are necessary in preparation for attending this course, those interested can consult the references listed below:

- A Model of Scientific Research in Psychology [4] is a good, Creative-Commons licensed book on research methods, freely accessible at <https://kpu.pressbooks.pub/psychmethods4e/chapter/a-model-of-scientific-research-in-psychology/>. I recommend reading part II, sections 7, 9, 10, 11. Each section is only a few pages.
- The original book on experimental rigor and validity is Shadish, Cook, and Campbell's *Experimental and Quasi-experimental Designs for Generalized Causal Inference* [3].

- For a deep dive into mediation and moderation, Baron and Kenney's original article *The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations* [1], is available, as are plenty of subsequent articles on the topic. We will touch on the concept of causal modeling, mediation, and moderation, but will not go into how to conduct mediation and/or moderation analyses.
- If you have not taken the human-participants research ethics training course, and plan to do research with human participants, you should enroll in the CITI Certification program or a comparable analogue. The CITI Program can be accessed at <https://about.citiprogram.org>.
- For those not familiar with writing academic articles, Daryl Bem's guide [2] is still quite good: <https://psychology.yale.edu/sites/default/files/bemempirical.pdf>. It's very sad that he ended his career by committing research fraud in his 'studies' of psi phenomena.

8 ACCESSIBILITY

For participants who have hearing impairments, a live transcription of the discussion can be provided if this is a virtual session or provided in the room if this is an in-person conference session. For participants with visual impairments, the slides and other materials can be provided in alternate formats.

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